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RESEARCH ARTICLE

An Integrated Preprocessing and Drift Detection Approach With Adaptive Windowing for Fraud Detection in Payment Systems

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ABSTRACT As fraudulent transaction methods evolve rapidly; it becomes progressively more challenging to detect them in payment systems. Static machine learning and rule-based traditional detection methods cannot capture all the dynamic and evolving nature of fraudulent behaviors, resulting in lower detection accuracy and a higher false positive rate. This study proposes a complete framework that brings together advanced data preprocessing, effective drift detection, and a reliable detection model to address these issues. The method uses Mutual Information and SelectKBest for selecting important features, applies ADASYN to handle class imbalance, and adopts Convolutional Neural Networks (CNN) to capture complex transaction patterns. By implementing Early Drift Detection Method (EDDM) and ADaptive WINdowing (ADWIN), the drift can be detected in advance and the system can respond to changes, both gradual and sudden. The framework was evaluated on three datasets, including real-world transactions and mixed-data environments, achieving superior accuracy, precision, and drift detection rates, with values up to 99.99% accuracy and 1.0 respectively. The findings show that the framework can adjust to changing patterns of fraud, reduce false positives, and enhance detection performance. These insights demonstrate the significance of dynamic pre-processing and drift-aware approaches in the context of real-time fraud detection. This also serves as a basis for future work in adaptive fraud detection model research areas such as of integrating online learning for improved speed and efficiency in high-frequency transactional environments.

INDEX TERMS Banking fraud, payment systems, prepaid cards, debit cards, credit cards, fraud detection, concept drift, class imbalance, data pre-processing, supervised feature selection, deep learning, drift detection, real-time fraud detection.

I. INTRODUCTION

Nowadays financial fraud has become one of the big issues faced by the banking industries [1]. Fraud can be defined

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as any activity carried out with the intent of gaining profit through deception. Basically, fraud is an illegal method for fraudsters to get funds and goods. Hence the goal of the fraudsters is to get the product without paying the amount or earning an unauthorized fund from an account. Technological advancement has led fraudsters to try new methods

to accomplish their goals. As a result, fraud incidents are increasing, therefore, the need to protect systems from fraud has become evident.

A payment system processes transactions between a customer (issuer) and a merchant (acquirer) without a complex process or cash, generally by cards or electronic money. Transactions fall into card-present (CP) requiring physical presence and input of PIN (e.g. ATM, POS) and card-not-present (CNP) occurring online (e.g., e-commerce). CNP transactions are a bit more prone to fraud since they can involve stolen card information. As payment volumes rise, so do instances of fraud [3].

Nowadays companies and banks are spending millions on Know Your Customer (KYC), Fraud Systems, and Anti Money Laundering (AML) systems, yet the number of fraudulent transactions is increasing [3]. Based on the Nilson Report, in 2019 there were gross fraud losses on Card-based payments which are around \$28.65 billion, amounting to 6.8¢ for every \$100 of total volume [3].

According to the Merchant Savvy website, global losses were 9.84\$ billion in 2011 and keep growing till it is expected to reach 40.63\$ billion in 2027 which is 25% higher than in 2020 [4]. Another statistic shared by the Merchant Savvy website is related to the methods of compromise used by fraudsters for different countries and the top value for each country was related to CNP transactions. Although there are many technologies designed and implemented to enhance fraud in payment systems, these techniques have not proven to be effective in protecting authorization and financial transactions from fraudsters. The reason is that the different types of fraud patterns are evolving which has a real impact on the implemented technologies to detect fraud [5].

Most of the banking industries use a rule-based approach to detect fraud in their payment system [6]. With rule-based fraud detection, the user usually configures a set of rules, such as account numbers, transaction types, amounts, etc. As part of the rule configuration, the user can also define which action the system should take if the transaction matches the rule. This can include declining the transaction, displaying an alert, blocking the card, or sending an SMS to the cardholder. Upon identifying the rule by the user, the rule becomes active, and fraud is detected when the transaction matches the rule.

Whenever there is a new suspicious transaction, the operator must verify whether it is fraudulent or otherwise and update the rules if necessary. The operator's verification process is time-consuming and costly for the organization. Occasionally, the operator might not check/miss some transactions until the bank or organization receives a fraud complaint from a customer. The fraudsters are, swiftly, and continuously adjusting their patterns and strategies to deceive.

The increase of e-commerce using various types of payment including prepaid cards, debit cards, credit cards, etc., is evolving fraud patterns so rapidly that the rules-based approaches cannot be updated fast enough or change quickly, leading to false positives (blocking good customers) and

inefficiencies in detecting fraudulent transactions [6]. Due to an increase of sophisticated fraud strategies, traditional systems no longer produce reliable results [6]. Thus, there is a need to address this problem.

The disadvantages of rule-based systems has motivated the researchers to turn to AI-based approaches as alternatives such as Machine Learning and deep learning [7]. Fig. 1 shows the difference between the rule-based and machine learning-based in the context of fraud detection systems. Rule-based systems depend on predefined rules. Therefore, these systems fail to reveal the latest patterns and its variants because they work on pre-defined rules. This limitation makes it difficult to address emerging tactics in the fraud space and the system performance may reduce when deals with rapidly evolving fraud tactics. However, machine learning techniques exploit data to learn autonomously and identify complex patterns, which improves the accuracy of fraud detection [6], [48]. Machine learning has emerged as a powerful tool for fraud detection, as these models can continuously adapt to new data and learn from subtle patterns and deviations [48].

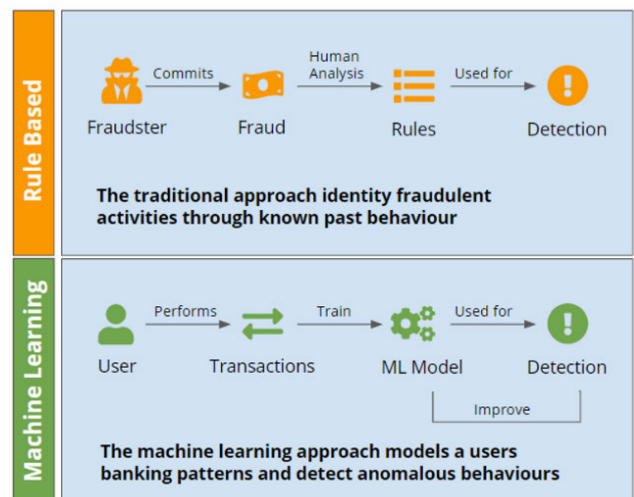


FIGURE 1. Behaviors in rule-based fraud detection system and ML fraud detection system [3].

There are different ML techniques proposed in different studies and each technique has its own advantages and disadvantages. ML techniques such as Support Vector Machines (SVM) are used in fraud detection, but they are not suitable for large data sets due to their high training complexity [8]. Hence with a large number of transactions such as in banking, this technique might not be practical to be implemented.

[9], [10] utilizes KNN for some of the analysis. The KNN requires clustering and would not work well with a large dataset since calculating distances between data instances would be very expensive. Furthermore, it is not effective when there is a high degree of dimensionality because the process of calculating distance for each dimension becomes complex. KNN is also sensitive to data that is noisy and missing [49]. Furthermore, it involves feature scaling, which

means all the data should be scaled (normalized, standardized) in the right way [9].

Data pre-processing is essential to prepare data in an appropriate format for machine learning applications. This step ensures data quality and accuracy before it is passed to the machine learning model, enabling the model to perform effectively. By refining the data through techniques like cleaning, normalization, and transformation, pre-processing enhances the reliability of the input data and significantly improves the model's detection performance.

Detecting payment card fraud is critical in the evolving landscape of payment systems. Several factors influence the effectiveness of fraud detection measures. One key challenge is the exponential growth in transaction volumes, which leads to more complex financial activities. Fraud detection systems must analyze these high volumes of transactions efficiently to differentiate between legitimate activities and fraudulent ones, a task that becomes increasingly difficult with the scale and complexity of modern payment systems.

Fraudsters exploit vulnerabilities in cards and merchant systems to carry out unauthorized transactions. While quick transaction processing is essential to ensure a smooth customer experience, it often compromises thorough verification, leaving systems exposed to fraudulent activities. Striking the right balance between fast processing and effective fraud detection is crucial. Robust fraud detection mechanisms must be designed to minimize fraud risks without impeding legitimate transactions, addressing this delicate trade-off in a scalable and efficient manner.

The problem characteristics of FDS can be described into two perspectives which are; challenges in fraud and limitations of the existing FDS solutions. One of the most significant challenges in fraud detection is the constant evolution of fraud strategies. Fraudsters continually adapt and modify their methods to bypass fraud detection systems (FDS). For example, a fraudster may steal a batch of credit card information and strategically use it for multiple smaller purchases across various online retail outlets. These transactions often mimic the cardholder's typical spending patterns, making it difficult for the FDS to identify them as fraudulent. This dynamic and ever-changing nature of fraudulent behavior highlights the need for adaptable and sophisticated detection mechanisms.

The main challenge with detecting fraudulent transactions is the class imbalance that is present in payment data, in which legitimate transactions far outnumber fraudulent transactions. The majority class can lead to bias in models as it becomes more dominant, and this limits the model's ability to detect less frequently occurred creative fraud activities. This works to study fraud detection systems and improving them.

Another significant trend in fraud detection is known as concept drift, which refers to the evolving nature of fraudulent behavior over time. In the context of payment card fraud, concept drift occurs when fraudsters develop new methods and tactics to evade existing detection systems. This evolution can be driven by factors such as the emergence of new

technologies, shifts in consumer behaviour, or the development of increasingly complex fraud strategies. Concept drift is not limited to specific timeframes; it can occur gradually over an extended period or as a sudden, abrupt change. This dynamic nature of fraudulent activity underscores the need for adaptive detection systems capable of keeping pace with these changes.

For example, in those types of systems a fraud detection model is being trained based on a dataset containing one year's worth of transactions. Models learn patterns such as large purchases, suspicious transactions, or high-frequency transactions. The system successfully flags transactions that exhibit these patterns as potentially fraudulent. However, over time fraudsters adapt and change their methods. To evade detection, they also find that smaller, less-obvious purchases or stolen credit card information to shop online may defeat the detection system. Hence, the pattern of fraudulent transactions are changing and a new set of features and rules may not be able to detect the different types of frauds.

To summarize, issues in fraud detection in payment systems can be categorized as follows:

A. HIGH OVERHEAD AND CLASS IMBALANCE

Many researchers have applied machine learning and deep learning to detect fraudulent payments, but their effectiveness is often hindered by redundant and irrelevant features, creating overhead. For example, features like exchange rates, billing currency, or account numbers are unrelated to fraudulent activity, as fraudsters typically exploit card numbers. Reducing this overhead through targeted feature engineering can significantly enhance detection accuracy and improve the efficiency of fraud detection systems (FDS).

Furthermore, card payment fraud is difficult to detect due to the highly imbalanced nature of the dataset [15]. As shown in Fig. 2, the credit card dataset contains a large number of positive instances (normal transactions) and only a small number of negative instances (fraudulent transactions) [17]. Due to this imbalance, classical classification methods perform poorly [16], as machine learning models are biased toward the majority class, which misclassifies the minority class (fraudulent transactions) [18]. This can be solved by reducing the data complexity prior to processing, which can be significant, allowing accurate and effective fraud detection.

Despite the highly imbalanced nature of Credit Card in which the positive instances make up only 0.17% of the total instances, the prediction is never 100% accurate due to the separability of features. Many features, in particular anonymized attributes (V1 to V28) have overlapping classes between fraudulent and non-fraudulent classes, resulting in infeasible model-perfectly distinguish classes. In general transactions, each fraudulent transaction shows similar feature distributions as that of legitimate transactions as they try to blend in with the normal activities in order to avoid being detected. Unfortunately, occasionally this overlap leads to misclassification of points, even with balanced data and

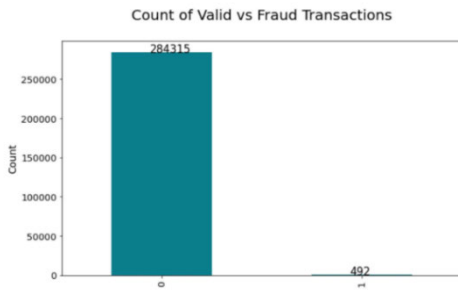


FIGURE 2. Among the credit card data datasets, 284806 transactions show statistics of 0.17% fraud (i.e., a small percentage of the transactions fall into the fraud class [40].

feature selection. In addition, adding noisy or non-informative features also impedes the ability to separate classes. These hurdles can be addressed with extensive preprocessing like data cleansing to eliminate noise/inconsistencies in the first step, balancing in the second and lastly feature selection to maintain only the most significant features. These measures greatly enhanced the model's capability to recognize distinguishing patterns and mitigate the effects of overlapping classes. This produced preprocessed datasets will be more suitable and it can be utilized for the later stage.

There are different studies [13], [27], [42], [43], [44] to tackle class imbalance. Some of them proposed to use over-sampling and some of them proposed under-sampling. A key difference between these two methods is that under-sampling moves instances from the majority class into the minority class to balance the data set while oversampling replicates the minority class instances to balance the dataset [19]. The drawback of each method is that under-sampling leads to the loss of potentially relevant information, causing the model to perform poorly while if the imbalance ratio is high, over-sampling introduces the disadvantage of overfitting and an additional computational cost [19].

Existing studies [2], [11], [13], [26], [27], [28] face limitations in pre-processing, often lacking thorough data cleansing, leading to incorrect AI model predictions and degraded performance. Some approaches use feature selection without clearly demonstrating the value or stability of the chosen features. Additionally, class imbalance handling is frequently applied after data preparation and feature selection, reducing the overall efficiency of the process.

B. HIGH FALSE POSITIVE

Customers' habits change over time, and so do fraudsters' strategies — this is known as Concept Drift [22]. The performance of FDS degrades when these patterns keep changing and evolving. Concept drift, a widely studied phenomenon in data streams, significantly impacts the performance of machine learning models, including fraud detection systems. It occurs when the statistical properties of data change over time, creating a mismatch between the model's training data and new incoming data. In credit card fraud detection, concept drift often arises from evolving fraud tactics,

shifting user behaviors, or changes within the payment ecosystem.

Concept drift and its effect on the effectiveness of existing fraud detection systems has been studied by several researchers. For example, [20], [21] showed that concept drift in credit cards can result in a higher rate of false alarms, and therefore the model fails to adapt itself to changes in fraud pattern in real-time. In another case [28], [47] reported lack of timely detection as well as handling stating that static fraud of concept drift compromised the fraud detection accuracy resulting in higher false positive rates.

Research highlights that concept drift is a persistent issue in credit card fraud detection, significantly contributing to increased false alarm rates over time. Studies [28], [29] show that static fraud detection models struggle to adapt to evolving patterns, leading to performance degradation. Lee [30] emphasized the importance of detecting and managing concept drift to preserve model performance and reduce false alarms. Addressing concept drift is essential for maintaining detection accuracy, minimizing operational costs, and improving user experience, making it a critical factor for effective fraud detection systems.

Fraudsters' strategies continuously evolve, making fraud detection systems (FDS) essential to detect new patterns promptly. As a result of static rules and outdated models that do not address the ever-changing nature of fraudulent behavior, current solutions have poor detection accuracy and high false alarm rates. In addition, the high operational overhead of large datasets with noise and nonfunctional features and with class imbalance degrades detection performance. Legitimate and fraudulent payment behaviors can shift significantly over time, requiring improvements in fraud detection systems. But current rule based and machine learning based methods cannot be readily adapted to cope with these challenges resulting in limitations in both accuracy and reliability.

To solve these problems, this paper uses a combination of improved processing, feature selection, adaptive resampling and hierarchical concept drift detection. The phase in pre-processing combines supervised feature selection techniques including Mutual Information with SelectKBest that will optimize pertinent features, as well as Adaptive Synthetic Sampling (ADASYN) to deal with the imbalance of classes. Furthermore, to keep it ahead of changes in fraud behavior a scheme is adopted which detects at micro level the process shift Adaptable Drift Detection (ADWIN) combined with Early Drift Detection Method (EDDM) for both smooth and sudden Changes in transaction pattern are implemented through a convolutional neural network (CNN), thus improving the accuracy of fraud detection. The main goal of the proposed way is to increase efficiency in identifying fraud, while also reducing false positives and ensuring that detection performance in constantly changing payment card transaction situations stays good.

This paper is organized into six sections. The first section introduces financial fraud, followed by the second section which gives an overview of related work. The third

section presents the proposed methodology including all pre-processing steps and concept drift detection approach. In this section also, the experimental setup, dataset used and detailed framework's technical implementation have been explained. The fourth section presents results and analysis. Meanwhile section five briefly describe future work and the final section concludes the paper, summarizing key findings, and highlighting the contributions.

II. LITERATURE REVIEW

This section explores the existing studies and works done by others on fraud detection in card payment systems. A brief overview of the card payment system and the most common fraud methods employed by fraudsters is provided at the beginning of this review. Following this foundational overview, existing research is analyzed in depth to identify fraud detection techniques, models, and methodologies. During this process, the review identifies research gaps and areas for further advancement.

A. PAYMENT CARD SYSTEM

A payment card is an electronic payment card that allows the cardholder to pay through POSs or withdraw money through ATMs, or e-commerce websites. Through a network of connections, card payment systems allow customers to make payments by card [35]. In the context of card payment, the payment ecosystem consists of the different players and their interactions that make the transaction work. The person using a payment card to purchase something or transact is known as the cardholder. Payment card issuer also known as Issuer, it is typically a bank or other financial institution that issues payment cards to cardholders. These cards are used by merchants (for example, retailers, or service providers) as a method of payment for goods or services. Acquirers are essentially banks or other organizations that partner with the merchants that accept and approve card payment. At the heart of this ecosystem are the networks, like Mastercard, Visa and American Express, which serve as intermediaries between issuers and acquirers. These networks are responsible for governing the cards' usage, laying the ground rules for where and how cards can be used, instituting fees and ensuring smooth, secure operations between parties. Fig. 3 shows how the payment card works over the network: A Customer performs a transaction on e-commerce, ATM or POS terminal at the merchant side. This transaction is then sent to the merchant through the payment system. The merchant terminal sends the transaction once again to the acquirer's payment system. The acquirer processes the transaction and forwards to the card network. It forwards transaction data to the issuing bank for approval or denial. The card network sends the response back to the issuing bank. The card network then routes the response back to acquirer, who would have either approved or rejected the transaction based on whether issuer approved or not.

There are three main types of payment cards: debit cards, credit cards, and prepaid cards. A **debit card** is linked

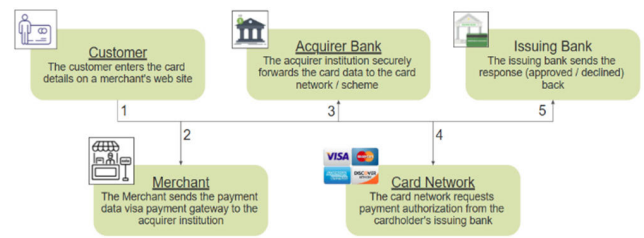


FIGURE 3. Payment card over the network.

directly to the cardholder's bank account, such as a savings or checking account. When the customer uses the card for transactions, the amount is immediately debited or credited to their bank account. A **credit card**, on the other hand, is linked to a shadow account, allowing the customer to borrow funds up to a pre-approved credit limit. The customer can use the card provided they pay their monthly dues. If payments are delayed, interest or late fees are charged. A **prepaid card**, also known as a pre-loaded card, is not tied to a bank account. Customers load money onto the card, which can then be used globally for various transactions such as cash withdrawals, purchases, or e-commerce. Unlike debit and credit cards, prepaid cards can be used by both customers and non-customers.

At the system level, payment processors play a critical role in facilitating transactions between issuers and acquirers. They format and process transactions, forward clearing information to merchants and card networks, validate transactions, regulate payment authorizations, and ensure the security of customer data against fraud and other threats.

B. EUROPAY, MASTERCARD, AND VISA (EMV)

EMV (Europay, MasterCard, and Visa) is a global standard for secure payment processing using smart card technology. It is named after the three companies that developed the initial chip technology embedded in EMV cards. These chips securely store and process cryptographic information during transactions, reducing fraud risk and ensuring global consistency in payment processing standards, terminal behavior, and communication protocols.

EMV transactions are categorized into two types:

1. **Contact Transactions:** The chip card is physically inserted into a terminal reader, such as an ATM, POS system, or other compatible devices.
2. **Contactless Transactions:** The card is tapped on a terminal's contactless reader, and data is exchanged via Near Field Communication (NFC).

An additional layer of security for contactless transactions is **tokenization**, which replaces sensitive card information with a unique token, further safeguarding the transaction from fraud [37].

C. TRANSACTION TYPES

There are two types of transactions which are card-present (CP) and card-not-present (CNP).

- CP transactions: transactions are those in which payment details are captured in person, at the time of sale. This occurs when cards are physically tapped via NFC, swiped over POS, or dipped into the terminal CHIP's reader to process transactions with the EMV [38]. Hence the Customer needs to physically insert or tap the Chip on the ATM for Cash Withdrawal, POS for purchase transactions, etc. [3]. The following are examples of CP transactions [39]: POS terminals with EMV Reader, Contactless terminals, CHIP card readers connected to smartphones
- CNP transactions: Cardholders who are not physically present when a transaction is conducted are considered Cards-Not-Present (CNP). In the present day, remote order transactions are the most common. This can be done over the phone, by fax, on the web, or by mail [38]. The CNP means is unlike Card-present transactions in which the transaction can be done online such as e-commerce transactions and in this kind of transactions, the fraudster always uses card details such as card number, expiry date, and CVV2 to perform transactions [3]. The following are examples of CNP transactions [39]: E-commerce, shopping carts, subscription or recurring billing, e-invoicing, ordering via phone and manual entry and the apps that allow mobile payments without requiring payment card readers.

D. FRAUD IN CARD PAYMENT SYSTEM

Payment card fraud is one of the challenges for businesses as it involves the illegal use of cardholder data to perform unauthorized purchases and transactions. They disguise unauthorized purchases and fake card as cardholder details, adapt their behavior and use fraud patterns that can circumvent security measures. Exploiting vulnerabilities in the payment system, such fraud results in financial losses and potential harm to both businesses and individuals.

[46] presented a few types of Fraud in Card Payment:

- i. Skimming: Using an electronic or manual imprinting tool, the fraudster can access the information displayed on the magnetic stripe. In other words, the fraudster can skim the card using different tools to read the magnetic stripe. Fraudsters may use this information in the future for fraudulent transactions if they gain access to it.
- ii. Card Not Present (CNP) Fraud: As previously explained, fraudsters can use a card without physically possessing it by entering card information such as the expiry date, card number, and CVV.
- iii. Fake card: Fake card fraud is usually done using skimming. Fraudsters make a clone mag stripe card with the same data stored on the original card. The card is a dummy but is still operational for future transactions.
- iv. Lost or Stolen card fraud: For e-commerce or manual entry transactions, fraudsters can use that card for transactions as no PIN verification is needed when the card is lost or stolen.

- v. Application Fraud: Application fraud is when someone steals payment card details, however, card ID theft is not to be confused with this. But fraudsters use the stolen card details to create a tenth account or make transactions. This type of fraud is hard to identify.
- vi. Fake Merchant Websites (Phishing): The victim of false merchant websites is similar to phishing fraud, where the customer is misled into a fake website created by fraudsters, which closely resembles a genuine site. To lure the customer into making a purchase, the fraudulent webpage may offer several discounts. When the transaction is completed, all the card and transaction information is collected by the fraudsters and used for fraudulent transactions.
- vii. Merchant Collusion: Merchant collusion occurs when the merchant forwards information related to the cardholder without the cardholder's knowledge.

The top three most common types of fraud:

- i. Skimming: This involves gathering information from a cardholder's magnetic stripe through magstripe reader or through other skimming tools. This data is also captured during valid transactions and is then used for subsequent illegal transactions. However, fraudsters can use skimming devices on payment terminals like ATMs to steal sensitive card information.
- ii. Card-Not-Present Fraud (CNP): CNP fraud is common in the online world. The thieves make illicit purchases online using the card number, expiration date, and CVV. The fact that e-commerce transactions do not require the presentation of a physical card may lead to them being also targeted by fraudsters, who can be definitively deceiving both financial institutions and merchants.
- iii. Fake Card Fraud: Fraudsters use stolen card information—often obtained through skimming—to create fake payment cards, which are then used for fraudulent transactions. The stolen data is encoded onto counterfeit cards, allowing fraudsters to make unauthorized purchases.

E. EXISTING WORKS IN FRAUD DETECTION

This section presents a review of the literature related to fraud detection systems (FDS) published from 2015 to 2024. The review discusses different approaches and methods to detect fraud done by existing works, highlighting the techniques used, limitations and their advantages. This section reviews related works about fraud detection in card payment systems and it focuses on five main challenges: class imbalance, concept drift, feature selection, classification techniques, and drift detection methods. The comparison of past approaches demonstrates their shortcomings and gives some insights on how to improve.

Shakya [24] examined the application of ML techniques in credit card fraud detection, using classification algorithms like Logistic Regression, Random Forest, and XGBoost.

This study utilized resampling methods (such as Random Under Sampling, Tomek Links Removal, SMOTE, and combinations of SMOTE and Tomek Links) to manage class imbalance, and the model achieved a high accuracy of approximately 98%. Shaky's research provided full experimental details for each machine learning that has been utilized demonstrating effective handling of class imbalance and robust classification performance. Fig. 4 demonstrates the workflow of this model:

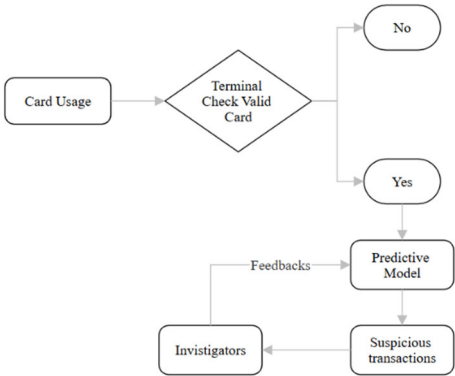


FIGURE 4. Fraud detection process flow in Shaky's [24] proposed model.

Evan [26] proposed a hybrid method combining J48 Decision Tree, SVM, K-means Clustering, and Random Forest as shown in Fig. 5. The approach involved clustering similar data points to detect anomalies and classifying transactions using Random Forest. Fraudulent transactions were flagged if identified by either model, while unclassified transactions were processed using confidence thresholds. The two-stage model increased discriminability and provided a robust detection, as evidenced by high classification accuracy and robustness.

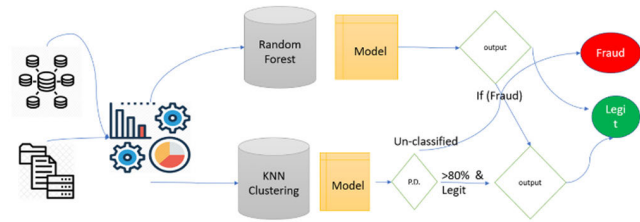
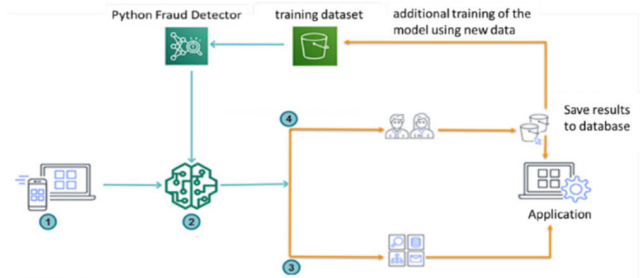


FIGURE 5. Proposed credit card fraud detection system by Evan [26].

In Shmatko's study [27], the Random Forest algorithm served as the primary classification method for credit card fraud detection. Transactions were assigned a risk score (0–100) based on a machine learning model trained on historical data. Transactions below a predefined threshold were classified as non-risky, while those exceeding the threshold (e.g., 90) were flagged for manual review. Bank staff could approve, reject, or further investigate flagged transactions, with outcomes used to update the dataset and retrain the

model for improved performance. This approach integrates manual oversight with model refinement, enhancing detection capabilities over time. Fig. 6 shows details of [27] proposed system.



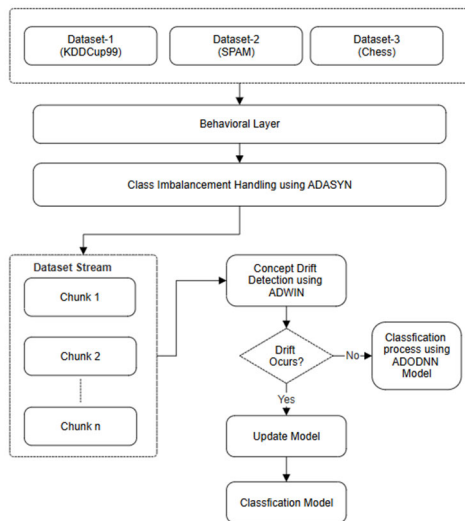


FIGURE 8. Workflow of CIDD-ADODNN model with ADASYN balancing and ADWIN drift detection [25].

imbalance in streaming data. The proposed model is illustrated in Fig. 8.

The model combines techniques for preprocessing, imbalance handling, drift detection, and classification. Preprocessing formats raw data, followed by ADASYN to balance class distribution, improving the detection of minority fraudulent transactions. ADWIN drift detection is then applied to identify and adapt to real-time shifts in fraud patterns, triggering self-updates to maintain high accuracy. The model employs a Deep Neural Network (DNN) optimized with Adadelata for precise transaction classification.

In 2017, S. Caxton Emerald and T. Vengattaraman [36] introduced a model named “Concept Drift Detection with Optimal Machine Learning Model for Data Classification.” Fig. 9 demonstrates the workflow of this model, starting with the input training dataset from the KDDCup99 and Spam datasets.

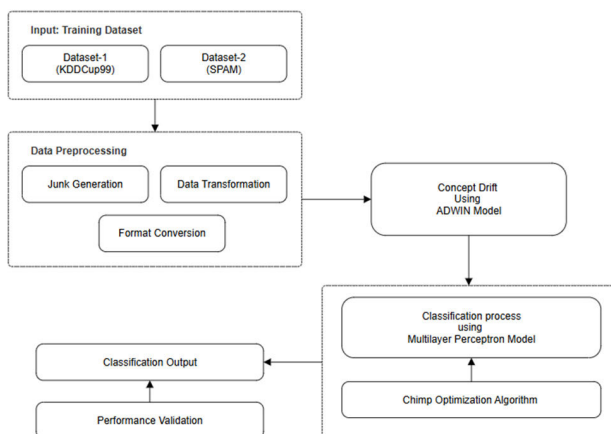


FIGURE 9. Workflow of the concept drift detection model with optimal machine learning for data classification [36].

The extracted raw content is then processed further such as chunk generation, data transformation and format conversion as a part of data preprocessing to make it ready to be analyzed. To detect such region shifts over time, the ADWIN (Adaptive Windowing) model is utilized. During data evolution, this step is essential to keep model performance.

As soon as drift detection is complete, the data is classified using a Multilayer Perceptron (MLP) model. The Chimp Optimization Algorithm further optimizes the model’s performance by fine-tuning the classifier parameters to produce more accurate predictions. Afterwards, the classification output is evaluated, and the model’s performance is validated.

The model’s strength is its incorporation of a drift detection mechanism. ADWIN can efficiently identify sudden drifts, allowing the model to adapt quickly to sudden changes in transaction patterns. The enhanced Chimp Optimization Algorithm ensures this model is also fine-tuned, paving the way for optimal performance by improving the classifier hyperparameter, thereby enhancing not only accuracy but also the adaptability to evolving trends in credit card fraud [36].

Reference [42] proposed an adaptive classifier framework to address concept drift and class imbalance in streaming data environments. The framework integrates drift detection, class imbalance handling, and classification into a cohesive workflow. It begins with preprocessing streaming data to prepare it for further analysis.

Drift detection is achieved using ensemble-based resampling and statistical methods like Kullback-Leibler (KL) divergence and Cramer’s V statistic to identify changes in data distribution. To address class imbalance, the framework employs adaptive oversampling of minority class instances. This ensures the model remains effective by retraining with augmented instances that reflect the latest data distribution. Classification is enhanced using robust ensemble methods such as Random Forest, Bagging, LightGBM, and XGBoost, which are particularly effective in handling noisy and complex data with outliers.

This approach offers improved adaptability and prediction accuracy, making it well-suited for dynamic and complex environments. However, the computational complexity of ensemble methods and adaptive resampling could pose challenges in high-frequency or resource-constrained applications.

The methodology proposed by Kajal and Kaur [43] addresses class imbalance in datasets through resampling techniques and feature selection. Preprocessing the data before applying classification algorithms combines Near-Miss undersampling with Information Gain. Feature extraction and balancing of the dataset are the first steps in the workflow.

In the proposed method, the Near-Miss undersampling technique is used to equalize the number of instances in fraud and non-fraud classes. As a result of Information Gain, the most relevant attributes are identified, thus reducing computational complexity and improving model performance. The

classification method was divided into the Naive Bayes and Decision tree algorithms with Decision Tree giving the better results. Using Information Gain the Decision Tree classifier with selected 18 features accuracy, precise, recall, F1 achieved at 97% This is much better than Naïve Bayes which gets 90% for all these metrics. The adoption of proposed methodology has been validated in context with European Cardholder dataset, which thus gives the potential of proposed methodology in prediction of better accuracies with respect to handling highly imbalanced data.

The model designed by Saraf and Phakatkar [44] is an ensemble model for fraudulent detection of credit cards. They consider class imbalances as well as changing fraud trends. Fig. 10 shows that the proposed bagging–boosting workflow improves the accuracy of different models. Data preprocessing involves the cleaning of data, outlier removal using the interquartile range (IQR) approach, and oversampling of the highly imbalanced datasets with the SMOTE technique.

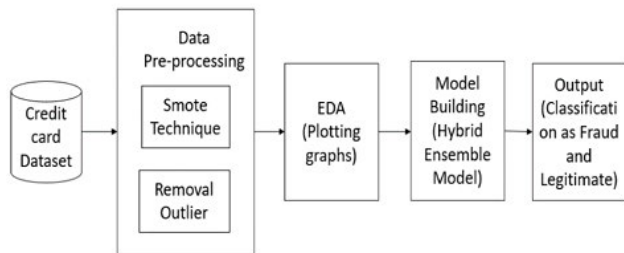


FIGURE 10. Workflow of the hybrid ensemble model for credit card fraud detection with smote balancing [44].

In classification phase hybrid ensemble, which combines Random Forest and AdaBoost. The Random Forest is an ensemble of decision trees trained on bootstrapped samples (with replacement) with an aggregated result, making it robust to overfitting, while AdaBoost performed on weighted voting focusing on misclassified instances to improve classification performance. Model with the hybrid ensemble outperformed Logistic regression Precision, Recall, and F1-Score are 1.00, 0.94, and 0.97 respectively by using the hybrid model, on the European credit card dataset, while on the Credit Card Stimulation dataset similar improvements are obtained. They described two datasets, the European dataset, and the stimulation dataset, that yielded precision-recall curves areas of 98.26% and 99.37%, respectively, supporting the model efficacy.

Similarly, Joshi et al. [51] proposed hard voting to identify fraudulent transactions by integrating multiple classifiers for credit card fraud detection using Decision Tree, Logistic Regression, and Naive Bayes. However, their model outperformed individual classifiers, and was able to detect fraud with higher accuracy. The study also highlighted that ensemble learning models can use different classifiers to overcome the difficulties of class imbalance. The caveat of this method is that it is not sufficient where there is concept drift, as it does not take into consideration the need for real-time adaptation

to changing fraud generative patterns. Joshi et al. [51] use architecture of ensemble learning-based credit card fraud detection model shown in Fig 11. It starts with the Preprocessing phase where transaction data in a raw form is cleaned up and processed. Since the dataset is heavily imbalanced, under sampling is used to reduce the number of data instances in the majority class, therefore making the dataset more balanced. The phase of Splitting Dataset slices the dataset into feature variables (independent) and target variables (dependent). Data ports from the Data Preparation to Apply Classifiers stage, where classifiers (e.g. Decision Tree, Logistic Regression, Naive Bayes) are applied. Apply Voting is a hard voting mechanism used to obtain the predicted class label, which is class with most predictions. Within the Evaluate Efficiency phase, some metrics (Accuracy, F1 score, Precision, and Recall) are used to evaluate how the model performs. This is an ensemble method, to augment the model (which might have had high class imbalances) on the top of classifiers capabilities.

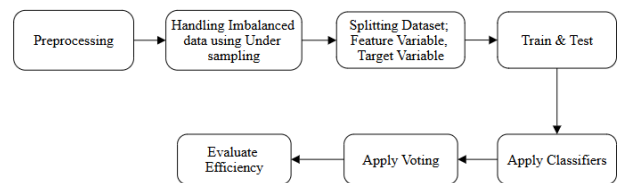


FIGURE 11. The process flow diagram for ensemble learning based credit card FDS. [51].

Singh et al. [12] used Random Forest classifiers for detecting fraudulent credit card transactions and addressed a class imbalance problem with Synthetic Minority Over-sampling Technique (SMOTE). Their approach provided a complete preprocessing mechanism including outlier removal, normalization, and feature scaling so that the model could be applied efficiently. The SHAP values variably represented the contribution of each feature to the model's output, which enabled the interpretable model and increased confidence in the model predictions. Singh et al. [12] to verify the relevance of individual features in fraudulent detection, making sure that the model not only attained great performance but also added insights for improving the system of fraud detection. The proposed method achieved 98.5% accuracy with very high precision and recall rates, also outperforming baseline models.

This literature review of fraud detection systems (FDS) between 2015 and 2024 discusses major contributions and denotes the gaps in existing models. While the reviewed papers addressed issues such as class imbalance, concept drift, and feature selection, it did not sufficiently enhance model adaptability and real-world applicability. Feature selection was neglected in many works so that the huge dimensions with irrelevant features (like Shakya [24] and Evan [26]) were increasing the computational cost. Methods like SMOTE and Tomek Links addressed class imbalance,

but introduced synthetic noise that led to overfitting and loss of generalizability. Furthermore, many models like Shmatko's [27] required manual feedback for data entry, hampering transaction processing speed and system adaptability as the project progressed.

Finally, concept drift is a persistent challenge in fraud detection given that transaction behavior is dynamic and can change with seasonality, new fraud techniques, and consumer behavior. The traditional models (e.g., Logistic Regression, Decision Trees, Random Forest, and XGBoost) assume that data distributions do not change over time. This assumption does not hold true for real-life payment systems, where a sudden or gradual drift can occur, significantly reducing the effectiveness of static models. Concept drift can be categorized into two types: abrupt drift, where data patterns change suddenly, and gradual drift, where transaction behavior evolves incrementally over time. For instance, seasonal spending variations or subtle shifts in merchant categories may gradually alter the decision boundaries of a model, making previously effective classification rules obsolete. Gradual drift is particularly problematic since it causes the performance of the detection to gradually decrease over time. As these changes happen unnoticed, the model keeps applying old rules and hence, a lot of false positives are left unnoticed leading to worse and worse accuracy. Models based on ADWIN [36] can detect abrupt drift without needing an initial reference set, and as such can be prone to lose gradual and subtle changes, causing missing patterns of potential frauds and degradation in model performance. Similarly, Mulimani's [42] approach, which integrated ensemble-based resampling with drift detection via KL divergence and Cramer's V, ensured robustness but was more computationally expensive, preventing it from obtaining real-time processing.

The study by S. Joshi et al. in [51] proposed an ensemble learning approach where Decision tree, Logistic regressor and Naive Bayes classifiers are applied independently and are then combined through hard voting to detect fraud in credit cards transactions. These results show that an ensemble method was found to yield higher accuracy as compared to any single classifier in prediction of fraudulent activity. The only limitation of this approach is that hard voting is used so no adaptive or incremental learning strategies for concepts drift modeling are utilized once new instances are coming. Furthermore, in dynamic fraud detection scenarios, ensemble learning techniques might be ineffective because they rely on a static model architecture and do not easily adapt well to changes in transaction behavior.

The model proposed by Singh et al. [12] has some limitations. Although SMOTE was able to solve the class imbalance problem, it also added synthetic noise to the dataset, which could increase the risk of overfitting to the synthesized dataset and result in poor generalization to real-world data. Furthermore, the model is unaware of concept drift, an important challenge in credit card fraud detection, as transaction patterns change over time. If no mechanisms for real-time

adaptation are embedded, the model's accuracy may degrade as new fraud patterns emerge. Furthermore, although SHAP values provided valuable insights into feature importance, the computational overhead associated with generating these values and performing extensive hyperparameter tuning may limit the scalability of the model, making it less practical for high-frequency transaction environments.

Table 1 summarizes these studies and their respective techniques as well as the strengths and limitations of the methods. These results show that, despite various methods to tackle class imbalance along with drift detection and feature selection, most of the models were not genuinely adaptive in real-time, nor did they accommodate the characteristics of high dimensional datasets or the changing nature of fraud.

Fortunately, the proposed model effectively identifies and reacts to gradual drift, which supports the mitigation of these effects. By perpetually adjusting to the changes in the transaction behavior, the model minimizes false positive rates while maximizing detection, mitigating the flaws of classical convergence and therefore, accommodating drifting detection models.

Drift is one of the most important factors that affect detection performance. Fraudsters regularly modify transaction characteristics (like transaction values, retailer types or geographical areas), moving the decision boundary of the prototype overtime. If no drift detection is in place, the models trained on historical data will become stale and inaccurate, which inherent in minimization of detection accuracy at higher levels specifically increase in false-positive and degradation detection performance. Now think about pushing this process one step back in the chain and start using drift-detection in a more proactive way so that you can make your models with dynamic transaction permutations to learn and adapt continuously. As an example, algorithms that can detect gradual drift, such as EDDM, can be integrated with popular algorithms to capture abrupt drift, such as ADWIN, so that both types of drift can be detected comprehensively. Deploying these strategies ensures that the model keeps pace with evolving fraud patterns, maintaining a balance between detection accuracy and adaptability. Previous models faced difficult challenges with the complexity of the dataset as many of them utilized numeric-only tabular datasets such as KDDCup and Credit Card datasets, which limited its applicability to real data, which is predominantly transactional and mixed (k+ numeric).

An ensemble approach was employed by Priya and Uthra [25] and Saraf and Phakatkar [44] to combine classifier outputs and increase classification accuracy but came with steep computational overheads and the potential for synthetic noise. Moreover, Kajal and Kaur's method [43] utilized the under sampling and Information Gain, but failed to address nonlinear feature interactions thus their solution is not flexible enough for real-world scenarios.

To overcome these limitations, a hierarchical feature selection process with Mutual Information and Feature Importance is applied to extract and rank relevant features, minimizing

the noise and improving classification performance. In addition, the proposed model is the first to combine EDDM and ADWIN to work together for dual drift detection to retain both sudden changes and gradual changes. This model can therefore adapt dynamically to drift of transactions and continues to perform detection at a high level. Additionally, the use of Convolutional Neural Networks (CNN) allows the model to recognize complex patterns in transactions, guaranteeing efficient, real-time fraud detection in ever-changing environments. The enhancements commence addressing the broader nature of fraud attempts and the limitations found on previous approaches, defined the proposed method as a more flexible and reliable way of acting against fraud.

III. METHODOLOGY AND THE PROPOSED MODEL

This section outlines an approach to developing an improved methodology to address limitations discussed in the previous section. Different challenges in fraud detection are handled in the research, like class imbalance, computational overhead and performance degradation owing to concept drift. It is important to go beyond these challenges for a fraud detection system to become more robust and effective.

This section describes the methodology used to develop and evaluate the proposed model that is capable of handling concept drift and imbalanced datasets. There were three distinct datasets utilized, each with unique characteristics relevant to fraud detection. To ensure effectiveness, accuracy, and robustness, multiple metrics were used to evaluate the model's performance. These steps are explained in detail as follows:

A. DATASET

The following datasets were selected to test the model's ability to detect fraud and adapt to drift in different contexts:

- i. Kaggle's first dataset contains European credit card transactions from 2013 and is commonly used in fraud detection research [50]. There are 31 features and 492 instances of fraud within 284,807 total transactions in this dataset, providing a benchmark for detecting fraudulent behavior under class imbalances.
- ii. Data from a leading bank in the Gulf Cooperation Council (GCC) region contains 200 features from debit and prepaid card transactions in 2019. This dataset provides insights into the model's performance in an operational banking environment with 120 fraudulent transactions out of 66,523 total records.
- iii. A third dataset, from the UCI Repository [45], contains labeled email data for spam detection research. 57 features were extracted from email text to provide a non-financial context for benchmarking concept drift handling. Using the spam dataset, the proposed drift detection techniques can be tested in scenarios of gradual and abrupt drift.

A summary of the datasets is shown in Table 2:

TABLE 1. Summary and comparison of fraud detection techniques in literature.

Ref.	Techniques	Strengths	Limitations
Shakya [24]	Logistic Regression, Random Forest, XGBoost; Resampling (SMOTE, Tomek Links)	Clear experimental procedures; handled class imbalance well; high accuracy (~98%).	Lacks feature selection; no drift detection; limited adaptability.
Evan [26]	J48 Decision Tree, SVM, K-means Clustering, Random Forest	Preprocessing includes cleaning and reduction; high accuracy with Random Forest (93.8%).	No feature selection, resampling, or drift detection; limited dataset and adaptability to evolving patterns.
Shmatko [27]	Random Forest	Straightforward workflow with client app integration for fraud scoring.	Missing feature selection, resampling, and drift detection; low accuracy (77%); lacks dataset diversity.
Yusof [11]	Fuzzy Association Rules, SVM, GRU	Comprehensive model with clear explanation of phases; includes behavioral analysis.	No drift detection or adaptive learning; lacks robust feature selection and undersampling details.
Taha & Malebery [13]	LightGBM, SVM, KNN, Logistic Regression, Random Forest	Incorporates feature selection and resampling; tested on two datasets; achieved high AUC (92.88%).	No drift detection or adaptive learning; limited adaptability in dynamic environments.
Priya & Uthra [25]	ADASYN for balancing, ADWIN for drift detection, ADODNN model	Handles class imbalance and drift detection; three datasets used; high accuracy across datasets.	Lacks feature selection; no adaptive or incremental learning.
D. Mulimani [42]	Ensemble-based Resampling, KL Divergence, Cramer's V, Random Forest, LightGBM, XGBoost	This technique detects drift very effectively; it is robust to class imbalance; it has been validated on multiple datasets; it can accommodate different types of drift.	The method is computationally intensive; there is a risk of synthetic noise during oversampling; the method is not generalizable to a wide range of datasets.
Kajal et al. [43]	Near-Miss Under-sampling, Information Gain, Decision Tree, Naïve Bayes	Effectively handles class imbalance; enhances feature selection; Decision Tree achieves 97% accuracy.	Data loss in Near-Miss under-sampling; overlooks nonlinear feature interactions.
Saraf & Phakatkar [44]	Random Forest, Adaboost (Hybrid Ensemble Model), SMOTE for resampling	Improves accuracy by combining bagging and boosting; effectively manages class imbalance; achieves an AUC of 99.37 percent.	The solution involves a high level of computational overhead, lacks explicit feature selection, and focuses primarily on drift detection.
Singh et al. [12]	Random Forest, SMOTE, SHAP for feature importance	Achieved 98.5% accuracy with detailed feature importance using SHAP values.	No drift detection; risk of overfitting due to synthetic noise; high computational cost for SHAP and hyperparameter tuning.
Tomar et al. [51]	Decision Tree, Logistic Regression, Naive Bayes, Hard Voting Ensemble	Higher accuracy compared to individual classifiers; mitigates class imbalance effectively.	No adaptive; limited adaptability to changing transaction behavior.

TABLE 2. Datasets specifications.

Dataset	European Cardholders (Kaggle)	Real Transactions (GCC Bank)	Spam Email (UCI Repository)
Availability	Public [50]	Private	Public [45]
Year	2013	2019	-
Features	31	200	57
Total Transactions	284,807	66,523	4,601
Fraud/Spam Instances	492	120	1,813 (Spam)
Noise	No	Yes	Yes
Class Imbalance	Yes	Yes	Yes
Drift	NA	Yes	Yes

The framework for the research is structured in two parts in which each stage builds on the findings of the last. Fig. 11 illustrates the designed fraud detection research framework.

- i. The first phase of data processing has three steps: data cleansing, feature selection and resampling. Data cleansing is the first phase of data quality in which data gets enhanced and corrected. Data cleansing is followed by features selection to select relevant features. Finally, The Resampling techniques has been utilized to solve the class imbalance problem and balance the dataset. Thus, combining these steps enforces efficient model training and detects fraud accurately.
- ii. The second phase deals with concept drift detection. The use of Convolutional Neural Networks (CNN), Early Drift Detection Methods (EDDM), and sliding windows allows the detection of both gradual and abrupt drifts in data.

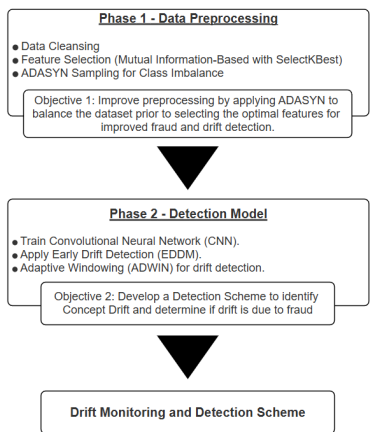


FIGURE 12. Proposed methodology for enhanced fraud detection with pre-processing and drift detection techniques.

B. METHODOLOGY

The methodology proposed consists of two phases: Data Preprocessing and Detection Model. The goal of each phase

is to tackle the major challenges of fraud detection, including data quality, relevant features, imbalance between classes and conceptual drift patterns.

1) PHASE 1 - DATA PREPROCESSING

The purpose of this phase is to prepare the dataset for effective and accurate model training. There are three steps involved in the data preprocessing: Data cleansing, feature selection and handling class imbalance with ADASYN.

Data Cleansing: This step aims to reduce noise and irrelevant information in a dataset to enhance data quality. Cleaning the data (missing values and duplicates) and thus, a dataset that can be analyzed. In payment systems which contain mixed types of data (i.e., categorical and numerical features), an integrated strategy is required during the balancing and feature selection process.

Handling Class Imbalance with ADASYN: ADASYN (Adaptive Synthetic Sampling) is a method used to deal with class imbalance, which is especially prevalent in fraud detection datasets:

- i generates synthetic instances of the minority class (fraudulent), focusing more on regions where learning is difficult. This helps in balancing the dataset by ensuring that areas with fewer but significant fraud cases are adequately represented.
- ii Adapts sampling to the underlying data distribution to enhance the model’s sensitivity to fraudulent transactions.

Thus, Phase 1 is focused on preparing a valid dataset to improve model accuracy, noise reduction and class imbalance correction using data cleansing, MI and K-Best feature selection and ADASYN balancing.

Feature Selection: To minimize the size of the dataset, Mutual Information (MI) and SelectKBest are utilized for better feature selection. MI measures the dependence between features and targets (fraud/non-fraud), producing scores that rank the features. SelectKBest then selects only those with the strongest association to the target variable, removing irrelevant or redundant ones. This results in a more accurate and interpretable model by eliminating low-impact features. Both methods are powerful for fraud datasets. LabelEncoder() is used for ordinal data, while OneHotEncoder() is applied to non-ordinal categorical data.

2) PHASE 2 - DETECTION MODEL

This phase focuses on building a dynamic model of fraud detection which would identify fraudulent transactions in real time with a high degree of accuracy, and most importantly a model that would look at data patterns that evolve due to changing tactics of fraud. 3 mechanisms are the building blocks for the detection scheme; they are Convolutional Neural Networks (CNN), Early Drift Detection Method (EDDM) and Adaptive Windowing (ADWIN).

CNN is further applied as a fundamental classification model to identify fraudulent transactions. This model

automatically identifies complex interrelations in transaction data which traditional techniques could miss. Training CNN on the Phase 1 dataset that is refined and balanced helps to distinguish real transactions versus fraud attempts better.

This method of monitoring the drift or change in the model's error on different upcoming batches of data based on the previous batches of data is termed as Early Drift Detection Method (EDDM) which can detect changes in distribution as well as fraudulent behavior. It constantly monitors the average distance between errors to pick up sudden or gradual changes in characteristics of transactions. It enables the system to respond quickly to new patterns in fraud by sending alerts when the error rate surpasses predefined warning and control thresholds.

The adaptive windowing (ADWIN) technique works along with EDDM to determine drift by adjusting the window size according to the volatility of recent transaction patterns. In ADWIN, the window size is dynamically adjusted to keep the detection model focused on current data trends, simplifying the process of detecting genuine shifts in fraudulent behavior from temporary abnormalities. Maintaining the model's accuracy and minimizing false positives is dependent on this component.

To address client-level drift and behavioral heterogeneity, the ADWIN component dynamically adapts its window size per client transaction stream, enabling localized detection of shifting behaviors. This ensures that the model remains sensitive to individual user-level changes in spending patterns, supporting personalized fraud detection in heterogeneous environments.

Fig. 12, shows a structured approach to fraud detection that integrates Convolutional Neural Networks (CNNs), Early Drift Detection Method (EDDM), and ADaptive WINdow (ADWIN) to continuously observe and adapt to changing transaction patterns. This framework accommodates both abrupt and gradual drifts in transactional data in order to maintain accuracy and be responsive to changing fraud tactics. In the CNN model, data such as transaction amounts, merchant details, and entry mode are extracted after initial training. The extracted features are used to detect anomalies and fraud.

The first step of the detection framework is Initial Model Training using CNN, where the CNN model is trained with historical transactional data, which learns complex rules that differentiate between normal and fraudulent transactions. This basic step guarantees that this CNN is implemented in real-time, meaning any incoming transaction can be dynamically classified as a baseline for fraud detection.

After the system is in production, the EDDM (Early Drift Detection Method) runs and monitors errors of the predictions in production in near-real-time. EDDM assesses for potential drift with two levels of thresholds: Warning Level (0.1) and Out-of-Control Level (0.1). If one of the significant deviations in the data patterns exceeds the out-of-control threshold, the EDDM detects the drift and classifies it as either Gradual Drift or Abrupt Drift. We identify a

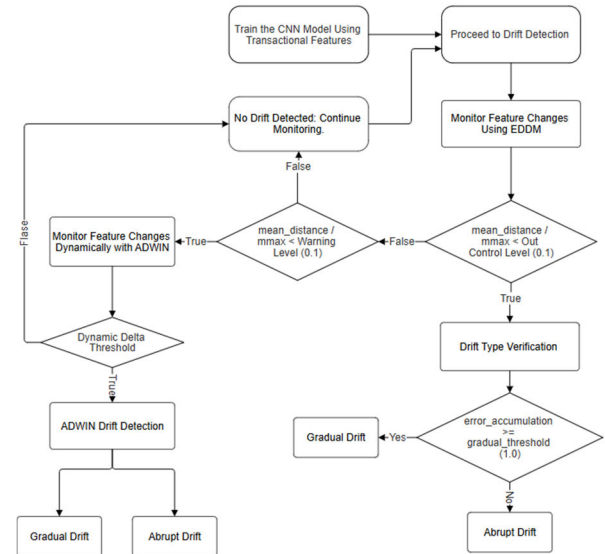


FIGURE 13. Drift detection Workflow in fraud detection system using CNN, EDDM, and ADWIN.

gradual drift if the error accumulation exceeds that gradual threshold, typifying gradual, incremental changes in the data distribution. In contrast, the drift in bringing to abrupt drift is sudden — and considerable. At this point, if a drift is detected, the system retries the CNN model or retrains its weights to adjust to the changes in the pattern.

If the Out-of-Control Level condition is False, the next to be evaluated is for the Warning Level. At this point, EDDM tends to monitor slight changes from the data stream. In case the warning condition holds True, EDDM triggers ADWIN (Adaptive Windowing) for further validation. ADWIN explicitly addresses this information gap by determining whether the observed shifts are true changes in data distribution or just temporary fluctuations. ADWIN confirms the drift type by adapting the size of its sliding windows dynamically and conducting statistical tests. In the occurrence of drift detection for the warning phase, ADWIN classifies the drift type within gradual or abrupt, the same as for the out-of-control phase. In the case where no drift is confirmed, the system reverts back to No Drift Detected and continues monitoring.

In case the warning level and out-of-control levels are not satisfied, the system will stay on tracking the data stream where it continues to have EDDM and ADWIN on standby to detect possible changes. This keeps it from generating false alarms while being sensitive to both subtle and significant drifts.

Upon confirmation of drift (EDDM or ADWIN), the system updates Drift Detection Metrics and implements necessary actions. In the case of gradual drifts, it updates its model weights to model the changing data according to the changes in this data. You are required to perform a complete retraining process with this CNN model every time it is a sudden drift, which we want to avoid as stated above. This

iterative process is essential for keeping the system flexible enough to respond to the evolving patterns of fraud.

Table 3 presents the main components and objectives of EDDM and ADWIN used in our approach for detecting drift and updating the model. When a change occurs, the system returns to the monitoring step and continues analyzing the data stream for further variations. This approach makes the system more reliable, keeps accuracy high, and allows it to adjust quickly to changes in fraud patterns.

TABLE 3. Summary of EDDM and ADWIN parameters for drift detection and model updates.

Parameter	Purpose	Role in CNN	Example	Impact
Warning Level	Early drift detection	Records drift without weight change	Small increase in high-value transaction	Heightened monitoring; stability maintained
Out-of-Control Level	Significant drift detection	Triggers model retraining/update	Sudden spike in fraudulent transaction	Immediate adaptation; reduces false negatives
Gradual Threshold (ADWIN)	Long-term drift adaptation	Adjusts weights gradually	Gradual rise in average transaction amount	Incremental updates; maintains model accuracy

C. EVALUATION METRIC

To gain an overall perspective about the model abilities, different metrics were used to be able to analyze the performance of the model. These metrics comprised:

- i. Accuracy: Accuracy is the proportion of correctly classified instances over the total number of instances. Drift detection in credit card or card payment fraud datasets requires high classification performance as transaction patterns evolve over time, making accuracy critical for successful detection. Nonetheless, even if the model is highly accurate, that means less in imbalanced datasets such as this one where the legitimate transactions outnumber the fraudulent ones. High accuracy only tells you that the model is broadly correct, but does not mean drift has been detected, nor that minority (fraud) examples are classified correctly. In addition, accuracy alone might not reflect the shifts in the distribution of data resulting from drift, and can cause the model to degrade if not detected in time.
- ii. Precision: Precision is the ratio of correctly identified positive observations to the total predicted positive observations. In drift detection uses cases; precision conveys that change in patterns occurring in fraud instances due to drift would not lead to nightmare-level false positives. As drift takes place, fraudsters may adapt to it by adjusting their behavior, causing transaction characteristics to differ. With high precision, the

- chance of false positives is decreased, meaning that fewer legitimate transactions will be flagged as fraudulent, reducing unnecessary alerts and disruptions.
- iii. Recall: Recall is the number of relevant instances (actual fraudulent transactions) retrieved by the model over the total number of instances that should have been retrieved (actual fraudulent transactions). So, in the case of drift detection, a high recall makes sure that the model adapts as quickly as possible to changes introduced by the concept drift, and do not let undetected fraudulent patterns manifest. Though drift changes the nature of fake transactions, having a high recall limits false negatives (there will be frauds they miss), ultimately holding the system safe from adapting threats. In drift scenarios, low recall means non-adaptation to changing patterns, hence the need for constant monitoring and retraining.
 - iv. F1 Score: The F1 Score is the harmonic mean of precision and recall, which balances the trade-off between false positives and false negatives. In drift detection, a high F1 Score indicates that the model has adapted to the changes in transaction distributions without sacrificing its flexibility to correctly identify fraud. Since both precision and recall get affected by drifting concepts, a high F1 Score guarantees that the model captures fraudulent transactions without hurting sensitivity or specificity along the evolution of the data.
 - v. ROC-AUC (Receiver Operating Characteristic - Area Under Curve): The ROC-AUC evaluates a model's capacity to separate fraudulent transactions from legitimate ones at various threshold levels. ROC-AUC is critical in drift detection as it serves to assess how well the model identifies a supervisory signal so that the classification boundaries shift dynamically with drift. This means the significantly high ROC-AUC value indicates the model's ability to determine between legalized transaction and fraudulent transaction as the characteristic of the transaction changes over a certain time. If the ROC-AUC remains consistently high, this indicates that even as drift affects the underlying data distribution, the model maintains its discriminative capability over time and is thus robust to such change.
 - vi. Drift Detection Rate: The Drift Detection Rate measures the model's sensitivity to detecting changes in the underlying data distribution over time. With the prevalence of emerging patterns amongst fraudulent transactions, detecting drift allows the model to retrain and adapt to changes in that process quickly. Without a high enough drift detection rate, the model can become outdated speculation, resulting in increased false negatives (missed fraud) or false positives (false alarms). Drift detection: The detection of shifts in the distribution of incoming data.
 - vii. Confusion Matrix: The confusion matrix divides result of classification into true positives, true negatives, false

positives and false negatives (TP, TN, FP, FN). The confusion matrix can be used for drift detection in fraud detection models to see how well the model is able to stay within class prediction accuracy as drift occurs. Tracking this matrix through time can show whether the model's sensitivity to changing patterns of fraud is getting better or worse, allowing for actionable insights on retraining and refinement.

The selection of these evaluation metrics was performed in order to keep high classification performance of the model while adapting to the concept drift. These metrics together provide an overview of the model's performance over time, focusing on how well the model is performing and how well it is adapting to changes as they occur.

D. IMPLEMENTATION DETAILS

Fraud detection using various kinds of python libraries tailored for machine learning, data preprocessing and drift detection were performed in google colab:

- i TensorFlow/Keras: used to create Convolutional Neural Networks (CNN) for investigating complex patterns in transaction data. CNNs constructs multiple layers: Conv1D, MaxPooling1D, Dense to learn complex patterns from large amounts of data.
- ii TensorFlow/Keras: used to create Convolutional Neural Networks (CNN) for investigating complex patterns in transaction data. CNNs constructs multiple layers: Conv1D, MaxPooling1D, Dense to learn complex patterns from large amounts of data.
- iii River: Used for drift detection Because River is designed for streaming data, this library is utilized for drift detection. River was used to implement the EDDM and ADWIN techniques, which perform well for sudden and gradual changes, respectively.

Table 4 outlines the main code components utilized in each stage:

IV. RESULTS AND DISCUSSION

This section provides the evaluation outcomes, where Real Dataset, the Credit Card Dataset, and the Spam Dataset were utilized in the study to evaluate the model's performance in detecting drift and fraud. Before performing the actual experiments, each dataset was optimized to improve the robustness and reliability of drift detection using ADASYN for class balancing. Then the features were extracted using Mutual Information and SelectKBest. Then the integrated approach of EDDM, ADWIN and CNN was used to evaluate the performance of the model as it drifted.

The findings illustrate the contribution of each feature set to the model's performance metrics, which include Accuracy, Precision, Recall, F1 Score, ROC AUC, Drift Detection Rate, and False Positive Rate. The evaluation highlights the most effective feature configuration for each dataset, resulting in the highest detection accuracy and the ability to adapt to changing fraud patterns.

TABLE 4. Main code components.

Step	Code Component	Description
Data Cleansing	drop_null_columns, replace_missing_values	Cleans the dataset by eliminating columns with null or zero values and imputing missing values according to their data type.
Feature Selection	SelectKBest, mutual_info_classif, Pipeline	Selects informative features, handles mixed data types, and uses MI-based ranking.
Drift Detection	Custom EDDM class, ADWIN from River	Detects drift with EDDM for early signs and ADWIN for continuous adaptation.
Model Training	CNN in Keras (Conv1D, MaxPooling1D, Dense)	Identifies relevant features, accommodates various data types, and employs mutual information for ranking.
Evaluation and Metrics	Scikit-Learn Metrics (precision_score, etc.)	Calculates Accuracy, Precision, Recall, F1, ROC-AUC, Drift Detection Rate, and Confusion Matrix.

A. REAL DATASET

Table 5 illustrates the results obtained over the Credit Card dataset applying to the ADASYN balancing technique and by carry on the feature selection process, and also compares the performance achieved by those predictive models that were built using different sets of four or fewer distinct features, accuracy, precision, recall and F1 here too are reported. The configuration with 90 features yielded the best performance metrics among those tested, featuring 100% accuracy, 100% precision, and 100% recall. This indicates that the underlying model is detecting fraud with world class sensitivity and little in the way of false positives or false negatives. The 10-features (99.99%) and 20-features (99.99%) datasets also performed very well, achieving high accuracy (99.98%) and perfect recall, respectively, indicating high efficiency in identifying a fraudulent transaction.

However, while fewer features performed well, they might not retain all important fraud-related patterns and this could affect generalizability. High scoring sets across every metric occurred at features 30, 50, and 80, indicating the optimal selections of features. 40, 60 and 70 features were also found that could be highlighted but they were not balance metrics as metrics were better for Precision and F1 scores. This indicates that although the majority of feature combinations used, after ADASYN and feature selection, provided highly positive results, some combinations—especially at 90 and above—buoyed an optimal, repeated, and good detection potential.

B. CREDIT CARD DATASET

The performance results of the Credit Card dataset post feature selection and data balancing is detailed in table 5. A 30-feature set with 99.88% accuracy, 99.76% precision, and 100% recall was the most balanced across all metrics

of the tested configurations. In the credit card dataset, the model accurately detected drift and fraudulent instances with minimal false positives and false negatives.

All other lower feature configurations (10, 15, 20 and 25 features) all scored well, getting between 0.9901 and 0.9975 for accuracy, and everything including the 10 features scored 0.9900 and above for F1. As such, the resulting 30-feature configuration is the optimal combination of accuracy and adaptability regarding drift and fraud detection, through features selection and balancing.

C. ESPAM DATASET

Table 5 presents performance results of Spam dataset; results after addressing data cleansing, feature selection and data balancing issues, to illustrate the impact on accuracy, precision, recall and F1 score. With 55 features, the results showed the most balanced performance with accuracy 0.9306, precision 0.9262, recall 0.9448, F1 score 0.9354 and ROC AUC 0.9706. These findings imply that it aids the model to identify spam instances with significant sensitivity as well as minimal false positives as well as minimal false negatives in these functions. Overall, other lower number of features also achieved good accuracy like 5 and 10 features but significantly lower precision and F1 score, indicating more false positive cases. The performance improvement is seen between 15 and 30 feature counts but the balance achieved by 55-feature configuration were not found in configuration with less features.

The 55-feature set is the best selection for the Spam dataset, providing a balanced method for correctly identifying both spam and non-spam instances with strong precision, recall, and overall accuracy. The chosen features provide strong detection efficacy while reducing errors, making it appropriate for dynamic data contexts.

D. DRIFT DETECTION RESULT

The final performance of drift detection was evaluated on the datasets using EDDM, ADWIN, and CNN methods after resampling and feature selection. Based on the results of this evaluation, the model has demonstrated its ability to adapt to evolving data distributions across a variety of datasets. The final performance of drift detection was evaluated on the datasets using EDDM, ADWIN, and CNN methods after resampling and feature selection. Based on the results of this evaluation, the model has demonstrated its ability to adapt to evolving data distributions across a variety of datasets. Table 6 shows the final performance result

According to Table 6, the feature selection analysis for the Real Dataset with 90 features, produced the best detection results with an accuracy, precision and recall rate equal to 100%. This demonstrates that the model is able to pick up on shifting transactional behaviors, whether gradual or sudden. For the Credit Card Dataset, the set of 30 features provided 100% accuracy and 100% F1 Score. This balance maintains high sensitivity while reducing false positives which is critical to fraud detection. A well-balanced precision and recall will provide a reliable model in a real

TABLE 5. Result of all datasets during pre-processing stage.

Dataset	Features	Accuracy	Precision	Recall	F1 Score	ROC AUC
Real Dataset	10	0.9999	0.9998	1	0.9999	1
	20	0.9999	0.9998	1	0.9999	1
	30	1	0.9999	1	1	1
	40	0.9999	0.9998	1	0.9999	1
	50	1	0.9999	1	1	1
	60	0.9999	0.9998	1	0.9999	1
	70	0.9999	0.9998	1	0.9999	1
	80	1	0.9999	1	1	1
	90	1	1	1	1	1
	100	0.9999	0.9998	1	0.9999	1
	110	1	0.9999	1	1	1
Credit Card Dataset	5	0.892	0.8911	0.8935	0.8923	0.9605
	10	0.9901	0.9806	1	0.9902	0.999
	15	0.9962	0.9924	1	0.9962	0.9996
	20	0.9966	0.9933	1	0.9966	0.9996
	25	0.9975	0.9951	1	0.9975	0.9998
	30	0.9988	0.9976	1	0.9988	0.9999
ESPAM Dataset	5	0.7945	0.8355	0.7642	0.7983	0.8728
	10	0.8639	0.8966	0.8411	0.868	0.9274
	15	0.8852	0.8876	0.898	0.8928	0.9508
	20	0.887	0.8971	0.8896	0.8934	0.9482
	25	0.8781	0.9009	0.8662	0.8832	0.9425
	30	0.8888	0.9308	0.8545	0.891	0.9571
	35	0.903	0.8976	0.9231	0.9101	0.96
	40	0.9119	0.9207	0.913	0.9169	0.9624
	45	0.9075	0.9458	0.8763	0.9097	0.9662
	50	0.919	0.9122	0.9381	0.925	0.9723
	55	0.9306	0.9262	0.9448	0.9354	0.9706

TABLE 6. Result of all dataset during detection stage.

Metric	Datasets		
	Credit Card	Spam	Real Dataset
Total Samples	568,628	5,617	13,2805
Accuracy	1.0	0.9934	1.0
Precision	1.0	1.0	1.0
Recall	1.0	0.9869	1.0
F1	1.0	0.9934	1.0
ROC AUC	1.0	0.9934	1.0
Drift Detection	0	0.9934	0
True Positives	284313	2792	66402
False Positives	0	0	0
True Negatives	284315	2788	66403
False Negatives	0	37	0

scenario where there is a need to identify fraudulent cases accurately but also limit false alarms. The Spam Dataset (55 features) achieved 99.34% accuracy, 100% precision, and 98.69% recall, specific to the variant. These findings validate the robust performance of the model to detect differences in datasets. In addition, the results demonstrate the applicability of the methodology in more general sense beyond financial fraud detection, as they highlight its capacity to cope with changing data distributions.

Figure 14 illustrates the drift detection overlap using Venn diagram on the Credit Card Dataset. As per the Venn diagram, the total of 284314 drifts detected on the Credit Card Dataset as seen from the Venn diagram in Fig. 10 EDDM detected 284,313 drifts, and 100% drifts was detected by EDDM. ADWIN only uniquely detected 1 drift, which was a gradual

one. There were 476 abrupt drifts and 283,838 gradual drifts. EDDM shows a high alignment with ADWIN indicating that EDDM captures all important drifts with only 1 missed drift (gradual). This demonstrates that EDDM is a sound method for capturing both the sudden changes and those that happen more gradually, in this data set.

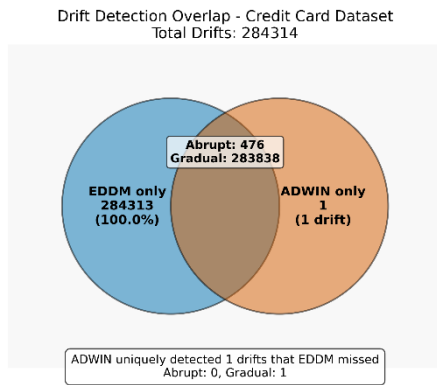


FIGURE 14. Drift detection overlap for the credit card dataset.

Figure 15 illustrates the drift detection overlap using Venn diagram on the ESPAM Dataset. As per the Venn diagram, the total of 2,792 drifts with EDDM capturing 100% of them. ADWIN did not identify any unique drifts, confirming that EDDM alone was capable of detecting all critical changes. The drifts included 75 abrupt drifts and 2,717 gradual drifts. These results emphasize the effectiveness of EDDM in identifying changes in this dataset, ensuring no drift instances were overlooked.

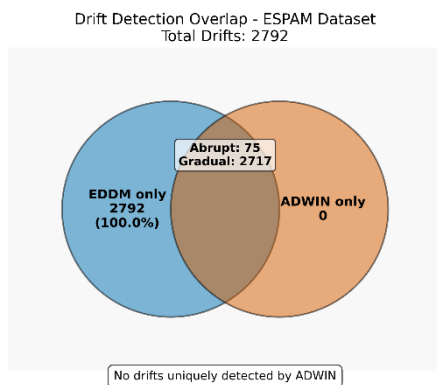


FIGURE 15. Drift detection overlap for the ESPAM Dataset.

Figure 16 illustrates the drift detection overlap using Venn diagram on the Real Dataset. As per the Venn diagram, the total of 66,402 drifts detected by EDDM. For this dataset, ADWIN did not identify any distinct drift as like for the ESPAM dataset. In summary, EDDM achieves 109 abrupt drifts and 66,293 gradual drifts. Such behavior substantiates that when functioning with real time events, EDDM is able toward detect drifts belligerently (observing no omission of any genuine drifts) which recommends its realism for genuine application where high compass for drifts is vital.

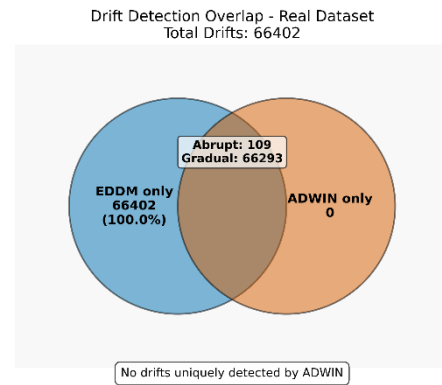


FIGURE 16. Drift detection overlap for the real dataset.

In most of the datasets, EDDM was able to successfully capture most of the drifts, including abrupt as well as gradual concept drifting. ADWIN, on the other hand, proved to find few drifts missed by EDDM and guaranteed that no major drift events went undetected. The complementing performance of both techniques demonstrates their robustness in achieving high detection sensitivity together. Utilizing both EDDM and ADWIN at different stages improves efficiency without sacrificing either speed or accuracy, making the approach more suitable for real-time environments.

While the dual-drift detector can enhance its adaptability, it does cause a small communication and computational overhead especially in the drift validation steps. Nevertheless, this is reduced by only updating the model where a drift has been confirmed.

E. PERFORMANCE COMPARISON

The performance of various drift detection and feature selection methods are displayed in Table 9 for the Credit Card dataset. When compared to earlier studies, the proposed model yields superior results in terms of accuracy, precision, and recall. Table 7 compares results from the Credit Card dataset with other studies demonstrating the proposed model enhanced detection of drift and classification performance.

In Table 8, the Spam dataset results are compared across studies, emphasizing the current approach's improved drift detection and classification performance. The Spam dataset was selected because it also exhibits concept drift, so it can be also part of evaluation to confirm the effectiveness of the technique using different dataset financial and non-financial datasets. By applying the proposed methodology to a non-financial dataset with drift, the ability of the proposed methodology to detect changes and maintain classification performance across different domains is validated. This highlights the adaptability of our approach to datasets where data distribution evolves over time.

In Table 9, the Real dataset's high results support the effectiveness of the proposed methodology. As a result, the model achieved near-perfect detection metrics, such as accuracy of 99.99%, precision of 99.99%, drift detection rate of 99.99%,

TABLE 7. Performance comparison on credit card dataset.

Credit Card Dataset	Kajal et al. [43]	Saraf & Phakatkarn [44]	Proposed Methodology
Dataset	Credit Card Dataset	Credit Card Dataset	Credit Card Dataset
Data Type	Only numeric	Only numeric	Only numeric
Techniques Used	Information Gain (K=18), Near-Miss Undersampling, Decision Tree	SMOTE for resampling, Hybrid Ensemble (Random Forest + AdaBoost)	Data Cleansing, ColumnTransformer, OneHotEncoder, LabelEncoder, Mutual Information + SelectKBest, ADASYN, CNN, EDDM & ADWIN
Drift Detection Rate	NA	NA	100%
Accuracy	97%	NA	100%
Precision	97%	100%	100%
Recall	97%	94%	100%
F1 Score	97%	97%	100%
ROC AUC Score	NA	98.26	100%
True Positives	NA	NA	284,313
False Positives	NA	NA	0
True Negatives	NA	NA	284,315
False Negatives	NA	NA	0

TABLE 8. Performance comparison on SPAM dataset.

ESPA Dataset	Emerald [36]	Mulimani [42]	Proposed Methodology
Dataset	Spam Email Dataset	Spam Email Dataset	Spam Dataset
Data Type	Only numeric	Spam Email Dataset	Mixed (Numeric + Categorical)
Techniques Used	MLP + ADWIN	Ensemble-based resampling + KL Divergence + Cramer's V	Data Cleansing, ColumnTransformer, OneHotEncoder, LabelEncoder, Mutual Information + SelectKBest, ADASYN, CNN, EDDM & ADWIN
Drift Detection Rate	NA	NA	99.34%
Accuracy	95.64%	98.87%	99.34%
Precision	NA	98.17%	100%
Recall	NA	98.29%	98.69%
F1	NA	98.21%	99.34%
ROC AUC	NA	NA	99.34%
True Positives	NA	NA	2792
False Positives	NA	NA	0 (0%)
True Negatives	NA	NA	2788
False Negatives	NA	NA	37 (0.6587%)

demonstrating exceptional robustness in handling complex, mixed data types.

F. K-FOLD CROSS-VALIDATION

The comparison results earlier showed that the proposed model achieves better performance than the existing methods.

TABLE 9. Result of detection stage using real dataset.

Attribute	Proposed Methodology
Total Samples	132805
Dataset	Real Dataset
Data Type	Mixed (Numeric + Categorical)
Techniques Used	Data Cleansing, ColumnTransformer, OneHotEncoder, LabelEncoder, Mutual Information + SelectKBest, ADASYN, CNN, EDDM & ADWIN
Drift Detection Rate	100%
Accuracy	100%
Precision	100%
Recall	100%
F1 Score	100%
ROC AUC Score	100%
True Positives (TP)	66402
False Positives (FP)	0 (0.00%)
True Negatives (TN)	66403
False Negatives (FN)	0 (0.00%)

However, it is also important to evaluate its stability when evaluated on various data subsets. K-fold cross-validation with K=5 was employed to accomplish this. Table 10 presents the outcomes across many test folds, indicating that accuracy, precision, recall, and F1-score remain consistent. The model's performance is not affected by how the data are split, indicating that it is reliable across different data distributions.

TABLE 10. Results using 5-fold cross-validation.

Datasets	K	Precision	Recall	F1-Score	Accuracy	ROC AUC
Real Dataset	1	1	1	1	1	1
	2	1	1	1	1	1
	3	1	1	1	1	1
	4	1	1	1	1	1
	5	1	1	1	1	1
Average		1	1	1	1	1
SPAM Dataset	1	1	1	1	1	1
	2	1	1	1	1	1
	3	1	1	1	1	1
	4	1	1	1	1	1
	5	1	1	1	1	1
Average		1	1	1	1	1
Credit Card Dataset	1	0.999991	1	0.999982	0.999991	1
	2	0.999991	0.999982	1	0.999991	1
	3	1	1	1	1	1
	4	1	1	1	1	1
	5	1	1	1	1	1
Average		0.999996	0.999996	0.999996	0.999996	1

Table 11 compares the proposed model's performance, evaluated using K-Fold Cross-Validation, with existing studies on the Credit Card and Spam datasets. The results indicate that the proposed approach achieves higher accuracy, precision, recall, and F1-score compared to previous methods. For the Credit Card dataset, the model recorded an accuracy of 0.99999, surpassing the 0.97000 reported by Kajal et al. [43]. As a result, the model effectively detected all drift

occurrences with a recall of 0.99999. Saraf & Phakatkar [44] reported a recall of 0.94, which indicates that certain distribution shifts may have been missed. Using the Spam dataset, the proposed model was able to recognize evolving patterns with an accuracy of 1.0. The precision and recall scores further highlight the model's capability to detect spam while minimizing false positives. These results reinforce the effectiveness of the proposed model across different datasets. The consistent performance across various K-Fold validation splits confirms its robustness and adaptability to different data conditions.

TABLE 11. Comparison of K-fold cross-validation and existing works.

Dataset	Metric	Proposed Model (K-Fold Avg)	Kajal et al. [43]	Saraf & Phakatkar [44]	Emerald [36]	Mulimani [42]
Credit Card	Accuracy	0.99	0.99	NA	NA	NA
	Precision	0.99	0.97	1	NA	NA
	Recall	0.99	0.97	0.94	NA	NA
	F1 Score	0.99	0.97	0.97	NA	NA
	ROC AUC	1	NA	98.26	NA	NA
Spam	Accuracy	1	NA	NA	0.9564	0.9887
	Precision	1	NA	NA	NA	0.9817
	Recall	1	NA	NA	NA	0.9829
	F1 Score	1	NA	NA	NA	0.9821
	ROC AUC	1	NA	NA	NA	NA

V. FUTURE WORK

Despite the good drift detection capabilities on various datasets of the current framework, it still does not support direct real-time learning based on changing fraud patterns. To solve this challenge, the adaptive learning model was developed in Phase 3, including the online learning mechanism with the SGD and a dynamic ensemble with LightGBM. This improvement allows the model to adapt to new transactional data incrementally without having to retrain over the whole dataset, which is crucial when deployed in high-frequency, concept-drifting applications like fraud detection. In the future, it is planned to investigate enhancements for this architecture using privacy-preserving methods (i.e., federated learning) to enable joint detection of fraud across institutions and protect sensitive data. Further, lightweight model architectures will also be explored to minimize computational and communication overhead to improve the responsiveness and scalability of the system in real-time applications.

VI. CONCLUSION

The present study presented an organized framework for drift detection and fraud identification, with performance evaluated via several datasets: Credit Card, Spam, and Real datasets. In summary, the incorporation of Mutual Information with SelectKBest feature selection, ADASYN for data balance, and advanced drift detection via EDDM and ADWIN provides a robust and flexible method for dynamic data environments. The proposed framework achieved 100% accuracy and a 100% drift detection rate on the Real dataset, demonstrating significant sensitivity and specificity for real-world fraud detection in both numeric and mixed-data contexts. This approach is obvious, comprehensible,

customized for real-time applications, in contrast to intricate ensemble models. The ability to adapt conducts future research, especially for integrating online learning to enhance adaptability to changing fraud patterns in both financial and non-financial sectors.

REFERENCES

- [1] A. K. Mishra, A. Pandey, and S. Anand. (Oct. 2019). *A Review on Credit Card Fraud Detection Using Machine Learning*. [Online]. Available: https://www.researchgate.net/publication/336552027_A_Review_On_Credit_Card_Fraud_Detection_Using_Machine_Learning
- [2] BitEffect. *Payment Systems: Principle of Operation and Opportunities*. [Online]. Available: <https://biteffect.net/payment-systems-principle-of-operation-and-opportunities/>
- [3] SDK Finance. *Detecting Payment Card Fraud with Machine Learning—H2O Driverless AI Kaggle Dataset*. Accessed: Dec. 14, 2024. [Online]. Available: <https://sdk.finance/detecting-payment-card-fraud-with-machine-learning-h2o-driverless-ai-kaggle-dataset/>
- [4] Merchant Savvy. (2023). *Payment Fraud Statistics 2023: Trends, Data, and Insights*. [Online]. Available: <https://www.merchantsavvy.co.uk/payment-fraud-statistics/>
- [5] H. Zou, "Analysis of best sampling strategy in credit card fraud detection using machine learning," in *Proc. 6th Int. Conf. Intell. Inf. Technol.*, New York, NY, USA, Feb. 2021, pp. 40–44. [Online]. Available: <https://dl.acm.org/doi/fullHtml/10.1145/3460179.3460186>
- [6] Mitek Systems. *How Does Machine Learning Help With Fraud Detection in Banks*. Accessed: Dec. 23, 2024. [Online]. Available: <https://www.miteksystems.com/blog/how-does-machine-learning-help-with-fraud-detection-in-banks/>
- [7] A. Arya. (7, 2024). *Fraud Detection Using Machine Learning Algorithms*. [Online]. Available: <https://intellipaat.com/blog/fraud-detection-machine-learning-algorithms/?U.S.#:text=Manual%20Review%20and%20Transaction%20Rules,-Nowadays%2C%20Machine%20Learning&text=Previously%2C%20industries%20were%20using%20a,detection%20to%20ML%2Dbased%20solutions>
- [8] D. Kumar. *Top 4 Advantages and Disadvantages of Support Vector Machine (SVM)*. Accessed: Dec. 23, 2024. [Online]. Available: <https://dhirajkumarblog.medium.com/top-4-advantages-and-disadvantages-of-support-vector-machine-or-SVM-a3c06a2b107>
- [9] A. Soni. *Advantages and Disadvantages of KNN*. Medium. Accessed: Dec. 28, 2024. [Online]. Available: <https://medium.com/anuuu.soni/advantages-and-disadvantages-of-knn-ee06599b9336>
- [10] S. Wandre, S. Desai, A. Patel, and H. Lopes, "Credit card fraud detection using KNN and naive Bayes algorithm," *J. Emerg. Technol. Innov. Res.*, vol. 9, no. 4, pp. 327–332, 2022. [Online]. Available: <https://www.jetir.org/papers/JETIR2204420.pdf>
- [11] T. A. Yusof. *Adaptive Model for Credit Card Fraud Detection*. Accessed: Dec. 23, 2024. [Online]. Available: https://www.researchgate.net/publication/339585842_Adaptive_Model_for_Credit_Card_Fraud_Detection
- [12] Z. Bawany and A. D. Shanbhag, "Using machine learning to detect credit card fraud," in *Proc. Int. Conf. Electr., Comput. Energy Technol. (ICECET)*, Cape Town, South Africa, Nov. 2023, pp. 1–7, doi: 10.1109/icecet58911.2023.10389421.
- [13] A. Taha and S. J. Malebary, "An intelligent approach to credit card fraud detection using an optimized light gradient boosting machine," *IEEE Access*, vol. 8, pp. 25579–25587, Feb. 2020, doi: 10.1109/ACCESS.2020.2971354.
- [14] BINUS. (2022). *The Importance of Data Preprocessing for Machine Learning in E-Commerce*. [Online]. Available: <https://sis.binus.ac.id/2022/07/11/the-importance-of-data-preprocessing-for-machine-learning-in-the-e-commerce-industry/#:text=In%20this%20case%2C%20data%20preprocessing,incorrect%20output%20from%20the%20AI>
- [15] R. A. Mohammed, K.-W. Wong, M. F. Shiratuddin, and X. Wang, "Scalable machine learning techniques for highly imbalanced credit card fraud detection: A comparative study," in *Proc. Trends Artif. Intell., Lect. Notes Comput. Sci. (PRICAI)*, vol. 11013, Jan. 2018, pp. 237–246. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-319-97310-4_27

- [16] D. Singh, S. Vardhan, and N. Agrawal, "Credit card fraud detection analysis," *Int. Res. J. Eng. Technol.*, vol. 5, no. 11, pp. 1600–1603, Nov. 2018.
- [17] T. Chioka. *The Class Imbalance Problem in Machine Learning*. Chioka's Blog. Accessed: Dec. 20, 2024. [Online]. Available: [http://www.chioka.in/class-imbalance-problem/#:~:text=What%20is%20the%20Class%20Imbalance,class%20of%20data%20\(negative\)](http://www.chioka.in/class-imbalance-problem/#:~:text=What%20is%20the%20Class%20Imbalance,class%20of%20data%20(negative))
- [18] R. Verma. *Class Imbalance: A Classification Headache*. Towards Data Science. Accessed: Dec. 20, 2024. [Online]. Available: <https://towardsdatascience.com/class-imbalance-a-classification-headache-1939297ff4a4>
- [19] M. S. Kraiem, F. Sánchez-Hernández, and M. N. Moreno-García, "Selecting the suitable resampling strategy for imbalanced data classification regarding dataset properties. An approach based on association models," *Appl. Sci.*, vol. 11, no. 18, p. 8546, Sep. 2021, doi: [10.3390/app11188546](https://doi.org/10.3390/app11188546).
- [20] A. Smith, B. Johnson, and C. Williams, "Concept drift and false alarm rates in credit card fraud detection," *J. Fraud Detect. Prevent.*, vol. 15, no. 3, pp. 178–187, 2021.
- [21] *Understanding Model Drift in Machine Learning*. [Online]. Available: <https://www.projectpro.io/article/model-drift-in-machine-learning/871>
- [22] A. D. Pozzolo, G. Boracchi, O. Caelen, C. Alippi, and G. Bontempi, "Credit card fraud detection: A realistic modeling and a novel learning strategy," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 8, pp. 3784–3797, Aug. 2017. [Online]. Available: <https://re.public.polimi.it/bitstream/11311/1044896/1/08038008.pdf>
- [23] A. Bifet and R. Gavaldá, "Learning from time-changing data with adaptive windowing," in *Proc. SIAM Int. Conf. Data Mining*, Jul. 2007, pp. 443–448.
- [24] R. Shakya, "Application of machine learning techniques in credit card fraud detection," M.S. thesis, Dept. Comput. Sci., Howard R. Hughes College Eng., Univ. Nevada, Las Vegas, NV, USA, 2018. [Online]. Available: <https://digitalscholarship.unlv.edu/cgi/viewcontent.cgi?article=4457&context=thesesdissertations>
- [25] S. Priya and R. A. Uthra, "Deep learning framework for handling concept drift and class imbalanced complex decision-making on streaming data," *Complex Intell. Syst.*, vol. 8, pp. 41–53, 2021, doi: [10.1007/s40747-021-00456-0](https://doi.org/10.1007/s40747-021-00456-0).
- [26] E. M. H. A. Rubaie, "Improvement in credit card fraud detection using ensemble classification technique," *Int. J. Nonlinear Anal. Appl.*, vol. 12, no. 2, pp. 1255–1265, Jun. 2021, doi: [10.22075/IJNAA.2021.5228](https://doi.org/10.22075/IJNAA.2021.5228). [Online]. Available: https://ijnnaa.semnan.ac.ir/article_5228_bffa45b11da19bafd8e8431ce1de1e05.pdf
- [27] O. Shmatko, V. Fedorchenko, and D. Prochukhan, "Detecting credit card fraud using machine learning algorithms," *InterConf*, vol. 71, pp. 393–403, Aug. 2021. [Online]. Available: https://www.researchgate.net/publication/354112967_DETECTING_CREDIT_CARD_FRAUD_USING_MACHINE_LEARNING_ALGORITHMS
- [28] S. Agrahari and A. K. Singh, "Concept drift detection in data stream mining: A literature review," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 34, no. 10, pp. 9523–9540, Nov. 2022. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1319157821003062>
- [29] C. Brown and L. White, "The impact of concept drift on fraud detection performance," in *Proc. ACM Int. Conf. Knowl. Discovery Data Mining*, Jun. 2017, pp. 256–268.
- [30] S. Lee, J. Kim, and M. Park, "Adaptive fraud detection using concept drift adaptationm," *IEEE Trans. Knowl. Data Eng.*, vol. 28, no. 9, pp. 2453–2467, Sep. 2018.
- [31] K. Kajal and K. Kaur, "Credit card fraud detection using imbalance resampling method with feature selection," *Int. J. Adv. Trends Comput. Sci. Eng.*, vol. 10, no. 3, pp. 1693–1699, 2021. [Online]. Available: <https://www.warse.org/IJATCSE/static/pdf/file/ijatcse801032021.pdf>
- [32] S. Mungase, A. Tikande, S. Bora, P. Zanwar, and A. Pathan, "Credit card fraud detection using machine learning framework," *Int. J. Innov. Res. Sci., Eng. Technol.*, vol. 7, no. 6, pp. 355–360, Jun. 2020. [Online]. Available: https://www.ijirset.com/upload/2020/september/104_Tejas_NC.PDF
- [33] E. Ileberi, Y. Sun, and Z. Wang, "A machine learning based credit card fraud detection using the GA algorithm for feature selection," *J. Big Data*, vol. 9, no. 1, pp. 1–18, Feb. 2022. [Online]. Available: https://www.academia.edu/76673761/A_machine_learning_based_credit_card_fraud_detection_using_the_GA_algorithm_for_feature_selection
- [34] R. Powar, R. Dawkhar, and P. Pratchi, "Credit card fraud detection using machine learning," *Int. J. Adv. Sci. Res. Eng. Trends*, vol. 5, no. 9, pp. 62–67, Sep. 2020. [Online]. Available: http://ijasret.com/VolumeArticles/FullTextPDF/546_9.CREDIT_CARD_FRAUD_DETECTION_USING_MACHINE_LEARNING.pdf
- [35] *Card Payments*. Accessed: Nov. 4, 2024. [Online]. Available: <https://www.psr.org.uk/our-work/card-payments/>
- [36] S. C. Emerald and T. Vengattaraman, "Concept drift detection with optimal machine learning model for data classification," in *Proc. 6th Int. Conf. Trends Electron. Informat. (ICOEI)*, Apr. 2022, pp. 1160–1165, doi: [10.1109/ICOEI53556.2022.9776949](https://doi.org/10.1109/ICOEI53556.2022.9776949).
- [37] S. Jain. *Guide To EMV—Contact & Contactless Payment*. Accessed: Dec. 20, 2024. [Online]. Available: <https://sruti-jain.github.io/img/GuidetoEMV-Contact-ContactlessPayments.pdf>
- [38] *What is a Card-not-present (CNP) Transaction and Why It Costs More*. Accessed: Dec. 31, 2024. [Online]. Available: <https://squareup.com/us/en/townsquare/what-is-a-card-not-present-transaction>
- [39] B. Dwyer. *What's the Difference Between Card Present and Card Not Present?*. CardFellow Credit Card Processing Blog. Accessed: Dec. 31, 2024. [Online]. Available: <https://www.cardfellow.com/blog/card-present-card-not-present-definition/>
- [40] A. Patel. (13, 2021). *Beginner's Guide To Classification Models: Catch Credit Card Fraud*. Codex. [Online]. Available: <https://medium.com/codex/beginners-guide-to-classification-models-catch-credit-card-fraud-fe5a73a3401f>
- [41] Cardinity. *Payment Flow*. Accessed: Dec. 31, 2024. [Online]. Available: <https://cardinity.com/support/payment-flow>
- [42] D. Mulimani, P. R. Patil, and S. G. Totad, "Adaptive classifier to address concept drift in imbalanced data streams," in *Proc. IEEE 2nd Int. Conf. Data. Decis. Syst. (ICDDS)*, Mangaluru, India, Dec. 2023, pp. 1–5, doi: [10.1109/icdds59137.2023.10434793](https://doi.org/10.1109/icdds59137.2023.10434793).
- [43] Kajal and K. Kaur, "Credit card fraud detection using imbalance resampling method with feature selection," *Int. J. Adv. Trends Comput. Sci. Eng.*, vol. 10, no. 3, pp. 2061–2071, May 2021, doi: [10.30534/ijatcse/2021/811032021](https://doi.org/10.30534/ijatcse/2021/811032021).
- [44] S. Saraf and A. Phakatkar, "Detection of credit card fraud using a hybrid ensemble model," *Int. J. Adv. Comput. Sci. Appl.*, vol. 13, no. 9, pp. 464–474, 2022. [Online]. Available: https://thesai.org/Downloads/Volume13No9/Paper_53-Detection_of_Credit_Card_Fraud.pdf
- [45] UCI Mach. Learn. Repository. (1999). *Spambase Dataset*. [Online]. Available: <https://archive.ics.uci.edu/dataset/94/spambase>
- [46] S. Saraf and A. Phakatkar, "Detection of credit card fraud using a hybrid ensemble model," *Int. J. Adv. Comput. Sci. Appl.*, vol. 13, no. 9, pp. 464–474, 2022.
- [47] B. Lebicot, G. Marco Paldino, G. Bontempi, W. Siblini, L. He-Guelton, and F. Oblé, "Incremental learning strategies for credit cards fraud detection: Extended abstract," in *Proc. IEEE 7th Int. Conf. Data Sci. Adv. Analytics (DSAA)*, Sydney, NSW, Australia, Oct. 2020, pp. 785–786, doi: [10.1109/DSAA49011.2020.00116](https://doi.org/10.1109/DSAA49011.2020.00116).
- [48] M. K. H. Chy, "Proactive fraud defense: Machine learning's evolving role in protecting against online fraud," *World J. Adv. Res. Rev.*, vol. 23, no. 3, pp. 1580–1589, Sep. 2024, doi: [10.30574/wjarr.2024.23.3.2811](https://doi.org/10.30574/wjarr.2024.23.3.2811).
- [49] A. Hachcham. (Jan. 27, 2025). *The KNN Algorithm—Explanation, Opportunities, Limitations*. [Online]. Available: <https://neptune.ai/blog/knn-algorithm>
- [50] Kaggle. (2016). *Credit Card Fraud Detection Dataset*. [Online]. Available: <https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>
- [51] P. Tomar, S. Shrivastava, and U. Thakar, "Ensemble learning based credit card fraud detection system," in *Proc. 5th Conf. Inf. Commun. Technol. (CICT)*, Kurmool, India, Dec. 2021, pp. 1–5, doi: [10.1109/CICT53865.2020.9672426](https://doi.org/10.1109/CICT53865.2020.9672426).



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